A Re-entrant Involution Engine and the Phantom Divide

From a Laws-of-Form kernel to a minimal quintom with an exact swap = negation symmetry

A combinatorial engine built from a fixed-point-free involution and a diagonal argument was examined for physical content. The engine carries no dimensions and cannot emit a number, so no cosmological parameter follows from it. One structural feature does transfer exactly. When the engine is equipped with the minimal action that lets it move, its defining involution becomes the swap of two scalar channels, that swap acts on the observable $p+\rho$ as negation, and its fixed point is the equation of state $w=-1$. The system then crosses the phantom divide repeatedly and, under Hubble friction, relaxes onto the fixed point it was combinatorially forbidden to occupy. This note records the kernel, the correspondence, the minimal model, and the numerical demonstration.

1 The formal kernel

The arena is the category of pointed involutive acts $(\alpha,\ a\colon \alpha\to\alpha,\ z)$ with $a\circ a=\mathrm{id}$, and morphisms the pointed equivariant maps. Three facts organize it.

1. $(\mathrm{Bool},\neg,\mathtt{true})$ is initial. The unique map out reads any object in one bit, and it is an isomorphism exactly when the act is a relabelled crossing, $\mathrm{Fund}(a)\iff a\cong\neg$. This is the law of crossing, $a\circ a=\mathrm{id}$, read as Spencer-Brown's mark.
2. Negation has no fixed point. By the Lawvere diagonal, for any self-reading $f\colon A\to(A\to\mathrm{Bool})$ the value $\mathrm{dodge}(f):=\lambda x.\neg(f\,x\,x)$ lies outside the image, so f is never point-surjective.
3. Feeding each diagonal back as a fresh point builds a strict tower, $L_{k+1}=\mathrm{Option}(L_k)$ with $\mathtt{none}\mapsto\mathrm{dodge}(f_k)$. Every stage embeds in the next and not back, the namer is never in the image of $\mathtt{some}$, and there is no rung below $L_0=\mathrm{Bool}$.

The re-entrant reading $S(a(c))=\neg S(c)$ never settles, which is the two-state oscillation Kauffman identifies with the imaginary value and the eigenform. The engine's motion is what it does in place of holding a value.

2 The one descriptor that transfers

Density evolves by the continuity equation

$$\dot\rho+3H(1+w)\rho=0,$$

so the dynamics turns entirely on the combination $1+w$. At $w=-1$ this vanishes and $\dot\rho=0$, so vacuum density is a fixed point of the dilution flow and the geometry is de Sitter. The value -1 is therefore a fixed point in cosmology for the same abstract reason it is the signature of a fixed-point-free involution in the kernel, and the two meanings coincide on one number without either producing it. The taxonomy of basepoints maps onto this directly: $a(z)=z$ (REST) corresponds to $w=-1$, and $x\neq\neg x$ (ESCAPE) corresponds to $w\neq-1$, where density genuinely evolves and the state carries an arrow.

3 Giving the engine dynamics

The single input added by hand is an action, since the combinatorics cannot supply one. On a flat FLRW background take two scalar channels, ϕ with a canonical kinetic term and σ with a phantom kinetic term, under a potential symmetric in the two:

$$\mathscr{L} = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}\dot{\sigma}^2 - V, \qquad V = \tfrac{1}{2}m^2\phi^2 + \left(V_0 - \tfrac{1}{2}m^2\sigma^2\right).$$

The offset V_0 keeps the energy density positive, and this is the minimal quintom, the smallest arena in which w can cross -1. Working in units $8\pi G = 1$, the densities are

$$\rho = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}\dot{\sigma}^2 + \tfrac{1}{2}m^2(\phi^2 - \sigma^2) + V_0, \qquad p = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}\dot{\sigma}^2 - \tfrac{1}{2}m^2(\phi^2 - \sigma^2) - V_0,$$

and the equations of motion, with the Friedmann constraint $H^2 = \rho/3$, reduce both channels to damped oscillators,

$$\ddot{\phi} + 3H\dot{\phi} + m^2\phi = 0, \qquad \ddot{\sigma} + 3H\dot{\sigma} + m^2\sigma = 0.$$

3.1 The involution becomes literal

The observable that decides everything is

$$p + \rho = \dot{\phi}^2 - \dot{\sigma}^2 = \rho\,(w+1).$$

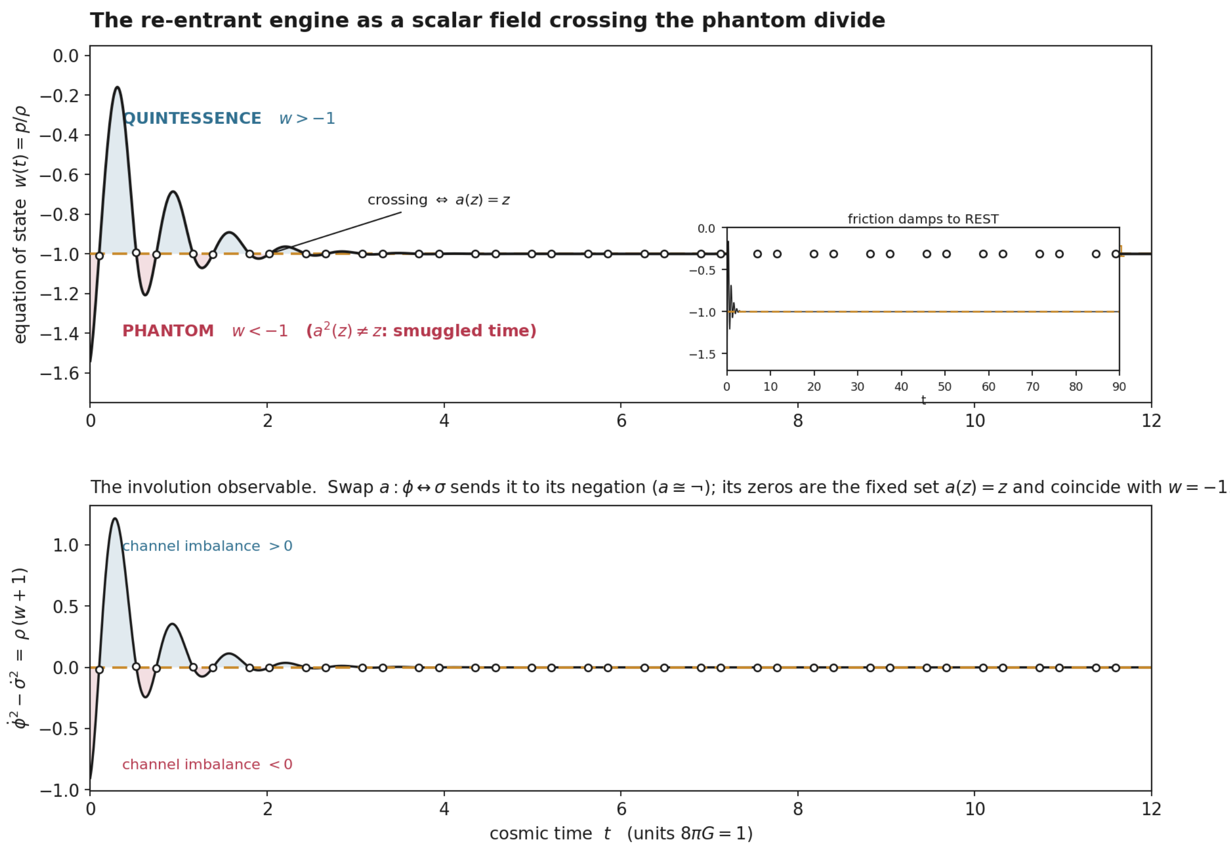
Let a be the channel swap $\phi \leftrightarrow \sigma$. It sends $p+\rho$ to $\dot{\sigma}^2 - \dot{\phi}^2$, its exact negation, so on the quantity measuring departure from the divide the swap acts as $\neg$, and $\mathrm{Fund}(a) \iff a \cong \neg$ is now a symmetry of the field equations. The fixed set $a(z) = z$ is $\dot{\phi} = \dot{\sigma}$, where $p + \rho = 0$ and $w = -1$. REST is de Sitter, seated at the fixed point of the swap.

3.2 The forced $+1$ is the second channel

A lone canonical field has $p + \rho = \dot{\phi}^2 \geq 0$, so it reaches $w = -1$ only as $\dot{\phi} \to 0$ and never passes below. Representing the crossing requires a strictly larger structure, the added phantom channel, so dodge forcing the next rung reads literally: the level below cannot name the crossing value, and the crossing demands $+1$. The phantom-divide no-go theorem for a single canonical field and the Lawvere diagonal are the same move.

4 Numerical demonstration

Integrating the system from unequal amplitudes with a phase offset, $\dot{\phi}^2 - \dot{\sigma}^2$ sweeps through zero every half period. In the window $t \in [0, 12]$ the equation of state crosses -1 thirty-seven times, swinging between phantom near -1.54 and quintessence near -0.16, while ρ stays positive throughout with minimum close to V_0. Every crossing lands on the swap's fixed set $\dot{\phi} = \dot{\sigma}$, which is why the zeros of the lower panel sit under the crossings of the upper one.



Upper: the equation of state $w(t)$ oscillating across the REST line $w = -1$, phantom below and quintessence above, with the inset showing relaxation onto $w = -1$ over many expansion times. Lower: the involution observable $\dot\phi^2 - \dot\sigma^2 = \rho(w+1)$, whose zeros are the fixed set of the swap $a \cong \neg$ and coincide with the crossings above.

5 What the dynamics adds, and what it does not

Hubble friction damps both channels, so the oscillation envelope $|w+1|$ collapses from order one to below 10^{-4} within a few expansion times, and the state relaxes onto $w = -1$. The pure combinatorial engine is excluded from its fixed point forever, and the field theory reaches that same fixed point asymptotically because it carries dissipation. This is the physical content that motion contributes, a mechanism that lets the re-entry decay, turning the excluded void into an attractor. The engine runs as the transient, and cosmic expansion is what finally lets it rest.

The boundary stays fixed. The parameters m, V_0, and the initial amplitudes are inputs, none of them descend from Bool, and the model forecasts no number. What it establishes is that the engine, once given an action, is a legitimate quintom that crosses the divide the current DESI data lean toward, with the correspondence exact where it is exact: swap $\cong$ negation, fixed point $\cong$ $w = -1$, and second channel $\cong$ the forced $+1$.